Spacetime Intervals, the Twin Paradox, and the Doppler Effect

PHYS 2203 — Introductory Physics III

September 30, 2026

Overview

We classify the separation between two events as **timelike**, **lightlike**, or **spacelike** according to the sign of the invariant interval — a frame-independent statement about whether one event can influence the other. That machinery settles the **twin paradox** posed in Lecture 11, once we work out the slope of a moving observer's *lines of simultaneity*. The spacetime diagram then shows exactly where the symmetry between the twins breaks, and the travelling twin really does return younger. We close with the **relativistic Doppler effect**, the change in the observed frequency of light when source and observer move apart or together.

1 Timelike, Lightlike, and Spacelike

Last lecture the 45° light lines through an event E divided spacetime into three regions: the **absolute future** of E, the events it can cause, the **absolute past**, the events that could have caused it, and the **absolute elsewhere**, the events no signal can connect to E at all and whose time order relative to E depends on the observer.

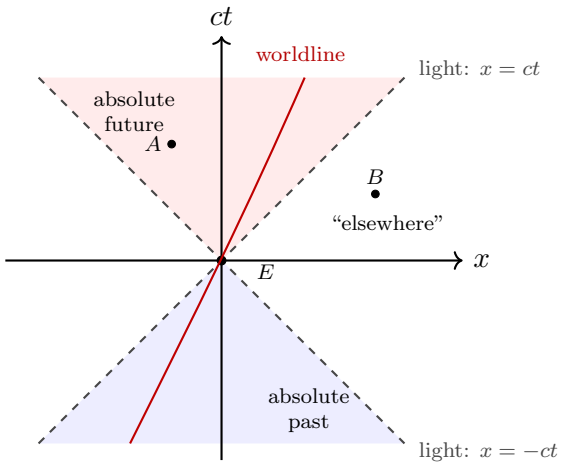


Figure 1: The light cone. The 45° light lines through an event E divide spacetime into the absolute future of E, the absolute past of E, and "elsewhere." Event A lies inside the future cone: it can be caused by E, and every observer agrees it happens after E. Event B lies outside the cone: no signal can connect it to E, and its time order relative to E depends on the observer.

Which region an event falls into is fixed by the *sign of the invariant interval* of Lecture 14,

$$(\Delta s)^2 = (c\,\Delta t)^2 - \left(\Delta x^2 + \Delta y^2 + \Delta z^2\right), \tag{1}$$

and because every observer computes the same $(\Delta s)^2$, every observer agrees on the classification. It is an absolute, frame-independent property of a pair of events.

Timelike, lightlike, spacelike. For two events separated by an interval $(\Delta s)^2$:

- $(\Delta s)^2 > 0$ — **timelike**. Inside the light cone. The events can be causally connected, their time order is the same for all observers, and there exists a frame in which they happen at the *same place*. The proper time between them is $\Delta\tau = \Delta s/c$.
- $(\Delta s)^2 = 0$ — **lightlike** (or *null*). On the light cone. Only a light signal can connect them, and $\Delta\tau = 0$.
- $(\Delta s)^2 < 0$ — **spacelike**. Outside the light cone, in "elsewhere." No signal can connect them, their time order is frame-dependent, and there exists a frame in which they are *simultaneous*.

2 Resolving the Twin Paradox

We are finally ready to settle the paradox posed in Lecture 11.

The setup. One twin, Anna, makes a round trip from Earth to a distant planet and back. The other twin, Bob, stays home. Anna travels at $v = 0.6c$, and the planet is 6 light-years from Earth as measured in Bob's frame. Do the twins age by the same amount?

The paradox. Each twin sees the other's clock running slow, so naively each should expect the other to be younger. They cannot both be right, since they stand side by side at the end and compare.

The two elapsed times

Bob's frame. Bob measures the proper distance 6 ly each way, covered at $0.6c$:

$$\Delta t = 2\,\frac{6\text{ ly}}{0.6c} = 20\text{ years}. \tag{2}$$

Anna's frame. For Anna the distance is length-contracted by

$$\gamma = \frac{1}{\sqrt{1-0.6^2}} = 1.25, \tag{3}$$

so she covers $L_0/\gamma = 6/1.25 = 4.8$ ly each way and her elapsed proper time is

$$\Delta t_0 = 2\,\frac{4.8\text{ ly}}{0.6c} = 16\text{ years}. \tag{4}$$

Result.

$$\boxed{\text{Anna returns } 20 - 16 = 4 \text{ years younger than Bob.}} \tag{5}$$

Lines of simultaneity: the slope of the x' axis

The diagram that resolves the paradox needs one more ingredient. A worldline records *where an object is at each time*. A **line of simultaneity** does something different: it records *everything that is happening at one instant*, according to a particular observer. "Now" is not a point — it is the whole of space at one moment — so on a diagram with one space axis it is a *line*.

For Bob, "now" means fixed t, so his lines of simultaneity are **horizontal**. They are his x axis and its parallels.

For Anna the answer is different. Her "now" is the set of events sharing a value of t', so set t' equal to a constant in

$$t' = \gamma\Big(t - \frac{v}{c^2}\,x\Big) = \text{const} \qquad \Longrightarrow \qquad t - \frac{v}{c^2}\,x = \text{const}, \tag{6}$$

and multiply through by c:

$$\boxed{ct = \frac{v}{c}\,x + \text{const.}} \tag{7}$$

Each choice of constant gives one instant of Anna's time, so her lines of simultaneity are a family of parallel lines of slope v/c. The member passing through the origin is her x' axis, and the others are its parallels. Bob's horizontal lines are the same result with $v = 0$.

The two axes of a moving frame. On a spacetime diagram drawn in Bob's coordinates, an observer moving at speed v has a worldline (her ct' axis) of slope c/v and lines of simultaneity (her x' axis) of slope v/c. The two slopes are reciprocals, so the axes are *mirror images about the* 45° *light line*, closing in on it from either side as $v \to c$.

Two important facts. First, the tilt carries the *sign* of v: reverse the direction of travel and the lines of simultaneity tilt the other way, just as the worldline does. Second, because Anna's "now" is tilted and Bob's is horizontal, the two of them slice spacetime into instants differently — which is the relativity of simultaneity of Lecture 11.

Where the symmetry breaks

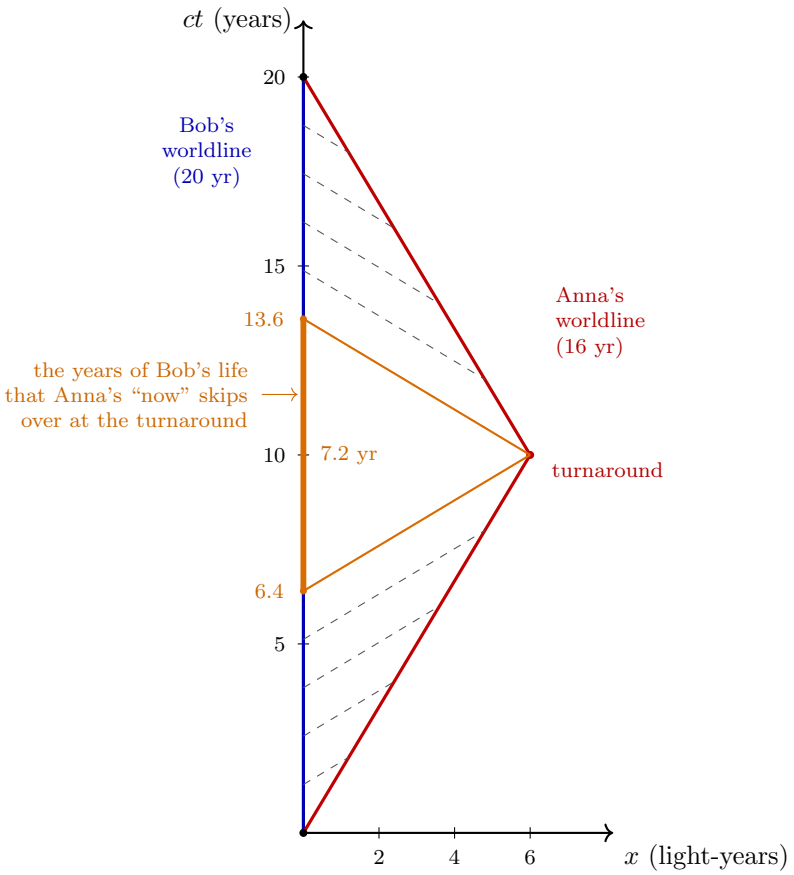


Figure 2: Spacetime diagram of the twin paradox. Bob's worldline is straight and vertical; Anna's is bent. The dashed lines are Anna's lines of simultaneity — the events she calls "now" on the outbound and inbound legs. At the turnaround her notion of "now" on Earth swings forward abruptly from 6.4 yr to 13.6 yr, skipping the 7.2 years of Bob's life marked in orange. Bob has no corresponding jump, and that is the asymmetry that resolves the paradox. The jump is a change in Anna's bookkeeping, not an event in Bob's life — nothing happens to him at that moment.

The diagram makes the resolution visible. The two worldlines connect the *same* two events — the departure and the reunion — but by different routes:

- **Bob's worldline is straight.** He remains in a single inertial frame for the whole trip.
- **Anna's worldline is bent.** She must accelerate to turn around, so she occupies *two* different inertial frames: one on the way out, another on the way back.

That bend is the whole story. Each twin does correctly observe the other's clock running slow *while both remain inertial*, and during those stretches the situation really is symmetric. But at the turnaround Anna changes frames, and with it her definition of which distant events are simultaneous with her "now." The orange lines in Figure 2 are her two lines of simultaneity at the turnaround, and they swing sharply upward as she reverses: a stretch of Bob's life that was still in her future on the outbound leg is already in her past on the inbound leg. The numbers are easy to get. The

outbound line through the turnaround event (6 ly, 10 yr) meets Bob's worldline at

$$ct = 10 - \frac{v}{c}(6) = 10 - 3.6 = 6.4 \text{ yr}, \tag{8}$$

and by symmetry the inbound line meets it at $20 - 6.4 = 13.6$ yr. So the band Anna's "now" skips over is

$$\boxed{\Delta t_{\text{skipped}} = 13.6 - 6.4 = 7.2 \text{ years} = \frac{2v\,\Delta x}{c^2}.} \tag{9}$$

Everything balances: Bob ages 6.4 yr during the outbound leg as Anna sees it, 7.2 yr during the turnaround, and 6.4 yr coming home — a total of 20 years. (And $6.4 = 8/\gamma$, which is just time dilation applied to each 8-year leg of her trip.) Bob, who never changes frames, experiences no such jump.

Resolution of the twin paradox. The twins are not interchangeable. Bob stays in one inertial frame, while Anna occupies two and must accelerate between them. Proper time is the "length" of a worldline through spacetime, and the bent path between two events carries *less* proper time than the straight one. Anna is genuinely younger when they meet.

3 The Relativistic Doppler Effect

We now derive the Doppler effect for light. Consider a source emitting wavefronts, and two observers.

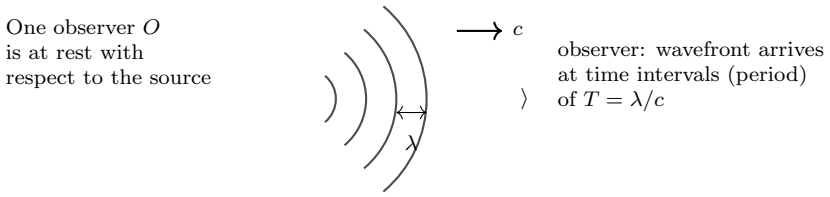


Figure 3: An observer at rest with respect to the source receives wavefronts of wavelength λ at period $T = \lambda/c$.

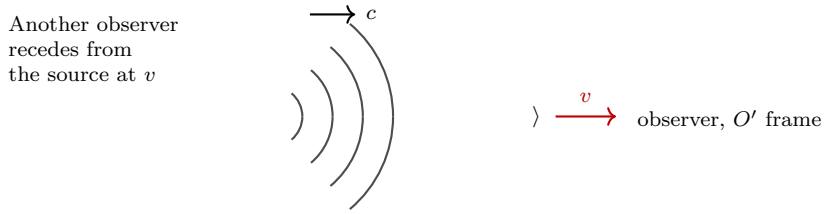


Figure 4: An observer receding from the source at speed v. Each successive wavefront must cover the gap λ *plus* the extra distance the observer gains in the meantime — the catch-up of Step 1. The conversion to the observer's own clock in Step 2 is where relativity enters.

The derivation takes two steps, and it is important to keep straight which frame each one happens in.

Step 1: the catch-up, in the source frame. Work first in frame O, where the source sits at rest and emits wavefronts a distance λ apart. The observer runs away at speed v, so each successive wavefront has to cover the gap λ *plus* the extra distance the observer gains in the meantime. If T_{rec} is the interval between arrivals *as measured in* O,

$$c\,T_{\text{rec}} = \lambda + v\,T_{\text{rec}} \qquad \Longrightarrow \qquad T_{\text{rec}} = \frac{\lambda}{c - v}. \tag{10}$$

This step is entirely classical — it is the same argument one makes for sound.

Step 2: convert to the observer's own clock. The two arrival events both happen *at the observer*, who is moving in O. So T_{rec} is a dilated interval, and the period the observer actually measures on their own wristwatch is the proper time

$$T' = \frac{T_{\text{rec}}}{\gamma}. \tag{11}$$

This — not any contraction of the wavelength — is where relativity enters. Combining the two steps with $\lambda = cT$,

$$T' = \frac{1}{\gamma}\,\frac{cT}{c - v} = \frac{T}{\gamma\,(1 - v/c)}. \tag{12}$$

Finally, writing $\beta = v/c$,

$$\gamma\,(1 - \beta) = \frac{1 - \beta}{\sqrt{1 - \beta^2}} = \sqrt{\frac{(1 - \beta)^2}{(1 - \beta)(1 + \beta)}} = \sqrt{\frac{1 - \beta}{1 + \beta}}, \tag{13}$$

so that

$$T' = T\sqrt{\frac{1 + v/c}{1 - v/c}}. \tag{14}$$

Frequency. Since $f = 1/T$, the frequency observed by an observer receding at v is

$$\boxed{f' = f\sqrt{\frac{1 - v/c}{1 + v/c}}} \;<\; f \qquad \text{(red shift, when receding)}, \tag{15}$$

where f is the frequency in the source frame and f' is the frequency in the frame of an observer receding at v.

Approaching source. For an approaching source, substitute $-v$ for v:

$$\boxed{f' = f\sqrt{\frac{1 + v/c}{1 - v/c}}} \qquad \text{(blue shift, when approaching)}. \tag{16}$$

Remarks.

* It does not matter whether the source recedes from the observer or the observer recedes from the source — the result depends only on their relative velocity. This is unlike sound, where the source's motion through the air and the observer's motion through the air give different formulas. Light has no medium, so only the relative velocity can appear.

* There is also a *transverse* Doppler effect, a purely relativistic shift observed when the source moves at right angles to the line of sight. It arises from time dilation alone and has no classical counterpart. We will not derive it here.